

**VIKASH VINOTH** | **Contact No.:** +1 940 629 7012 | **E-Mail:** [vikash.vzf34@gmail.com](mailto:vikash.vzf34@gmail.com)

OSS Solution Architect | Network Management | System Integration | NOC Automation | Fault & Inventory Management | Service Assurance & Fulfilment | Project Management | CI/CD Pipelines | Network Optimization | Pre-sales & Solution Design | Quality Assurance | Optical Transmission Networks | Client Interaction | SLA Compliance | GCP-AWS Cloud-based Solutions

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### **Professional Summary**

Experienced OSS Solution Architect with over 15 years in network management, system integration, and project leadership. Proven ability to design, implement, and manage Optical Transmission Networks, ensuring seamless system operations and automation. Skilled in fault management, inventory, performance, service assurance, and network optimization. Expertise in leading cross-functional teams, managing complex projects, and handling client interactions, from pre-sales to post-sales support. Adept in managing Scrum processes, testing phases, and CI/CD pipelines. Currently pursuing a Master's in Data Science to enhance technical expertise.

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### **Professional Experience**

**NMSWorks Software** - Chennai

*Oct 2013 – Jul 2024*

**Role:** OSS Solution Specialist Apr-2020 – Jul-2024

#### **Product & Backlog Management**

- Managed product backlog, roadmap, and feature prioritization.
- Collaborated with stakeholders to align requirements with business goals.
- Defined and communicated product priorities to development teams.
- Conducted UAT, reviewed deliverables, and ensured quality standards.

#### **Solution Engineering & OSS Implementation**

- Designed and delivered scalable, cost-effective technical solutions.
- Mapped customer requirements into functional and non-functional specifications.
- Led end-to-end OSS solutions for new product integrations (OEMs & third-party services).
- Managed FCAPS solutions for Ciena, Nokia, Cisco, Infinera, ASN & SubCom.

#### **Technical Integration & Protocol Handling**

- Integrated diverse NBIs: REST API, gRPC, CLI, SNMP, CORBA, ISN-IOO.
- Worked with authentication modes: Token-based, Credential-based, Certificate-based.
- Collected and analysed Inventory, Fault, and Performance dumps.

#### **Service Assurance & Fulfilment**

- Managed Fault, Performance, Trouble Ticketing, Provisioning, and Inventory.
- Led planning, optimization, and configuration for network operations.

#### **Pre-Sales & Project Delivery**

- Handled RFP preparation, evaluation, price negotiation, and requirement analysis.
- Delivered technical presentations, feasibility studies, and risk assessments.
- Owned technical solutions in pre-sales, ensuring smooth transition to operations.

#### **Infrastructure & Security Management**

- Automated deployments and ensured scalable CI/CD pipelines.
- Managed secure infrastructure, including servers, databases, and cloud migrations.
- Monitored system performance and implemented security best practices.

#### **Documentation & Stakeholder Management**

- Created solution architecture, design documents, and technical specifications.
- Provided customer feedback for product enhancements and business analysis.

**Role:** Lead System Engineer Apr-2017 – Mar-2020

- Product Owner of OSS tool BOSE, managing Google's Subsea Network Infrastructure.
- Developed a multi-vendor, multi-domain, multi-tenant OSS solution.
- Managed key modules, including Service Fulfilment, Service Assurance, Ticketing, Order Management, DCN Management, Spare Management, and IP Address Management.
- Led New Product Integration (NPI) for SubCom's TEMS and ASN's SN10.
- Worked extensively on Ciena Waveserver Ai, WS-5, 6500, RLS, and Nokia PSI-M & PSI-8L.
- Integrated AIS tracking in OSS for ship monitoring.
- Developed automated network failure notifications via SMS, WhatsApp, Email, and Auto-Ticketing.
- Designed network adaptors for OEM NBI communication.
- Led pre-sales activities, including RFPs, SRS, solution architecture, and compliance checklists.
- Conducted product demonstrations and proof-of-concept testing.
- Provided inventory solutions for complex networks, including ASON in metro and long-distance networks.
- Acted as Project Coordinator, bridging gaps between users and product delivery teams.
- Delivered product training for Network Operations Center (NOC) users.
- Integrated third-party ticketing, inventory, and customer support portals with OSS.
- Specialized in Subsea projects, including PFE, Wet ROADM, LME, Repeaters, and BU.
- Implemented Spectrum Sharing solutions using wave portal for Open Cable Interface (OCI).

**Role:** Senior System Engineer Oct-2013 – Mar-2017

- Conducted requirement gathering workshops and analysed User Business Requirements (UBR).
- Analysed North Bound Interface (NBI) data across SDH, EoSDH, WDM, OTN, MPLS, and PTN domains.
- Worked with WDM/OTN OEMs (Huawei, Ciena, Alcatel, NEC, ADVA, Cisco, Infinera) and SDH OEMs (ECI, Ciena, Fibcom, Alcatel, Huawei, Marconi, Tejas).
- Managed various NBI interfaces, including Open Config (gRPC/gNMI), REST API, CORBA/TMF-814, SNMP, CLI, and XML.
- Designed logic to interpret raw NBI data for inventory modelling in OSS.
- Developed Root Cause Analysis (RCA) logic for network alarm correlation.
- Created user stories and functional test cases for inventory, fault management, and RCA.
- Led User Acceptance Testing (UAT) and secured user signoffs.
- Authored solution design documents and test cases.

**Projects:**

- Tata Communications Ltd, Bharti Airtel Ltd, Idea Cellular Ltd and Seacom

**Bharti Airtel Ltd** - Chennai/Gurgaon

*Jun 2009 – Oct 2013*

**Role:** Network Operations Engineer

- Managed real-time operations, network expansion, re-engineering, and optimization.
- Expertise in SDH, DWDM, and OTN technologies.
- Worked with NMS platforms: ECI V5.0, Alcatel RM1354, Nortel, Huawei, and Tellabs.
- Led network expansion activities: Card addition, Optical patching, Network element addition, and Optical handovers.
- Executed network re-engineering: Re-parenting, Card/Element removal, Optical patching removal, and Capacity upgrades.
- Performed network optimization: Ring upgrades (STM4 to STM16, STM16 to STM64) and Traffic migration.

- Provisioned and shifted circuits (E1, DS3, STM1–STM16, Ethernet over SDH).
- Handled bandwidth upgrades, diversity provisioning, and customer traffic outage resolution.
- Ensured SLA adherence by providing Root Cause Analysis (RFO) post-restoration.
- Coordinated with internal teams to enhance network stability and minimize outages.
- Managed Change Management processes, approving operations after network health checks.
- Monitored alarms, reported issues to Network Surveillance, and generated MIS reports.
- Managed Inventory & Order Management using Metasolv M6.
- Hands-on experience with ticketing tools: HP OpenView & BMC Remedy.

**Projects:** Led award-winning project for diversity of mobility circuits, enhancing 100k mobility circuits.

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### Technical Skills

- **Network Management:** Ciena BluePlanet NCS/MCP/One-Control, ASN SN10, Huawei U2000, Alcatel 1353/54 NM/RM, ECI LightSoft
- **OEM:** Ciena, Nokia, Cisco, Huawei, ECI, Tejas, Fibcom, Marconi, Nortel, Alcatel, Tellabs
- **Telecom Technology:** SDH, DWDM, OTN, IP/MPLS, PTN, Ethernet over SDH
- **Database:** MySQL, Oracle, Cloud SQL
- **Cloud Platforms:** AWS, GCP, Azure
- **OSS Modules:** Fault Management, Inventory, Service Assurance & Fulfilment, Performance Management, Trouble Ticket, GIS, Spare Management, Document Management, AIS Ship-Tracker, Order Management
- **North Bound Interfaces:** RESTAPI, CORBA/TMF-814, SNMP, CLI, XML,
- **AAA Protocols:** Radius, TACACS+,
- **Bug-Tracking / Ticketing Tools:** BMC Remedy, JIRA, HP-Open View, Redmine
- **Project Management:** Agile, Sprint, Backlog, Change Request, RACI Matrix, Capacity Planning, Gantt Chart, Scrum
- **CI/CD & Testing:** Test Design, Test Execution, Defect Management, Source Control, Build/Release Pipelines
- **Inventory Management:** Cramer, Metasolv M6
- **Operating Systems:** Windows, Linux
- **Dashboard:** Tableau, Power BI
- **Language:** Python

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### Education

- **Master of Science in Data Science (Pursuing – Dec-2025)** – University of North Texas, Denton
- **Bachelor of Engineering** – Electronics and Communication Engineering, Anna University, Chennai (74.18%)

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### Key Projects

- **Google Subsea Network Management:** Led the design and implementation for optical network management.
- **Bharti Airtel Network Diversity:** Managed diversity across 100,000+ mobility circuits, awarded for excellence.
- **Tata Communications & Idea Cellular:** Handled large-scale network operations and optimization.